

Experiment No. 1

Encryption and Decryption Program in C

Program in C:

```
#include
<stdio.h> int
main()
{
    int i, x;
    char str[100];

    printf("\nPlease enter a
    string:\t"); gets(str);

    printf("\nPlease choose following
    options:\n"); printf("1 = Encrypt the
    string.\n");
    printf("2 = Decrypt the
    string.\n"); scanf("%d", &x);

    //using switch case statements
    switch(x)
    {
    case 1:
        for(i = 0; (i < 100 && str[i] != '\0'); i++)
            str[i] = str[i] + 3; //the key for encryption is 3 that is added to ASCII value

        printf("\nEncrypted string: %s\n",
        str); break;

    case 2:
        for(i = 0; (i < 100 && str[i] != '\0'); i++)
            str[i] = str[i] - 3; //the key for encryption is 3 that is subtracted to ASCII value

        printf("\nDecrypted string: %s\n",
        str); break;

    default:
        printf("\nError\n");
    }
    return 0;
}
```

Output:

Please enter a string: hello
Please choose following
options:
1 = Encrypt the string.
2 = Decrypt the
string. 1

Encrypted string: khood

Please enter a string: khood
Please choose following
options:
1 = Encrypt the string.
2 = Decrypt the
string. 2

Decrypted string: hello

Experiment No. 2(a)

Enter Email ID & Password

Program in C++:

```
#include <iostream>

using namespace std;

class LoginManager{
public:
    string userNameAttempt;
    string passWordAttempt;

    LoginManager(){
        accessGranted = 0;
    }

    void login(){
        cout << "Hey you need to enter your password and
        user.\nUsername:"; cin >> userNameAttempt;
        if(userNameAttempt==userName)
        { cout << "Password:";
          cin >> passWordAttempt;

          if(passWordAttempt==passWor
          d){

              cout << "Hey, that's right.";

          }
        }
    }
private:
    string passWord = "easypassword";
    string userName =
    "user@email.com"; bool
    accessGranted;
};

int main()
{
    LoginManager
    loginManagerObj;
    loginManagerObj.login();
}
```

Output:

```
Hey you need to enter your password and
user. Username:user@email.com
Password:easypassword
Hey, that's right.
```

Experiment No. 2(b)

Enter Username & Password

Program in C:

```
#include <stdio.h>
#include <string.h>
#include <conio.h>

//created by

chaitanya int main()
{
    char username[15];
    char password[12];

    printf("Enter your username:\n");
    scanf("%s",&username);

    printf("Enter your password:\n");
    scanf("%s",&password);

    if(strcmp(username,"manishdabhade")==0){
        if(strcmp(password,"admin")==0){

            printf("\nWelcome.Login Success!");

        }else{
            printf("\nwrong password");
        }
    }else{
        printf("\nUser doesn't exist");
    }

    return 0;
}
```

Output:

```
Enter your
username:
greatapple598
Enter your password:
admin
```

```
Welcome.Login Success!
```

Experiment No. 2(c)

Hiding Password

Program in C++:

```
#include
<iostream>
#include <string>

using namespace std;
void
    askForSecretPassword(voi
    d){ string password;
    cout<<"Enter your password: ";
    cin>>password;
    int
    len=password.length();
    system("cls");
    cout<<"Enter numeric password:
    "; for(int i=0;i<len;i++){
        cout<<"*";
    }cout<<endl;
};
int main(){

    askForSecretPassword();
    system("pause");
    return 0;
}
```

Output:

Enter numeric password: *****

Experiment No. 3 (a)

Caesar Cipher

Program in C++:

```
#include <iostream>

using namespace std;

string encrypt(string text, int s) {
    string result = "";
    for (int i = 0; i < text.length(); i++) {
        if (isupper(text[i]))
            result += char(int(text[i] + s - 65) % 26 + 65);
        else
            result += char(int(text[i] + s - 97) % 26 + 97);
    }
    return result;
}

int main() {
    string text = "ATTACKATONCE";
    int s = 4;
    cout << "Text : " << text;
    cout << "\nShift: " << s;
    cout << "\nCipher: " << encrypt(text, s);
    return 0;
}
```

Output:

```
Text : ATTACKATONCE
Shift: 4
Cipher: EXXEGOEXSRGI
```

Experiment No. 3 (b)

Substitution Technique

Program in Python:

Python program to demonstrate Substitution Cipher

```
import string
```

A list containing all characters

```
all_letters = string.ascii_letters # a-zA-Z
```

Create a dictionary to store the substitution for the given alphabet in the plain text

```
dict1 = {}
```

```
key = 4 # Substitution shift key
```

```
for i in range(len(all_letters)):
```

```
    dict1[all_letters[i]] = all_letters[(i + key) % len(all_letters)]
```

```
plain_txt = "I am studying Data Encryption"
```

```
cipher_txt = []
```

Loop to generate ciphertext

```
for char in plain_txt:
```

```
    if char in all_letters:
```

```
        temp = dict1[char]
```

```
        cipher_txt.append(temp)
```

```
    else:
```

```
        cipher_txt.append(char)
```

```
cipher_txt = "".join(cipher_txt)
```

```
print("Cipher Text is:", cipher_txt)
```

Create a dictionary to reverse the substitution

```
dict2 = {}
```

```
for i in range(len(all_letters)):
```

```
    dict2[all_letters[i]] = all_letters[(i - key) % len(all_letters)]
```

Loop to recover plain text

```
decrypt_txt = []
```

```
for char in cipher_txt:
```

```
    if char in all_letters:
```

```
        temp = dict2[char]
```

```
        decrypt_txt.append(temp)
```

```
    else:
```

```
        decrypt_txt.append(char)
```

```
decrypt_txt = "".join(decrypt_txt)
```

```
print("Recovered Plain Text:", decrypt_txt)
```

Output:

Cipher Text is: M eq wxyhCmrk Hexe IrgvCtxmsr

Recovered plain text : I am studying Data Encryption

Implementation of DES Encryption

Program in C:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    int i, ch;
    char cipher[50], plain[50];
    char key[50];
    int lp;

    while(1)
    {
        printf("\n__Menu__\n");
        printf("1. Data Encryption\n");
        printf("2. Data Decryption\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &ch);
        getchar(); // Clear newline character from input buffer

        switch(ch)
        {
            case 1:
                printf("\nData Encryption");
                printf("\nEnter the plain text: ");
                fgets(plain, sizeof(plain), stdin);
                plain[strcspn(plain, "\n")] = 0; // Remove trailing newline

                printf("Enter the encryption key: ");
                fgets(key, sizeof(key), stdin);
                key[strcspn(key, "\n")] = 0;

                lp = strlen(key);
                for(i = 0; plain[i] != '\0'; i++)
                    cipher[i] = plain[i] ^ lp;
                cipher[i] = '\0';

                printf("The encrypted text is: ");
                for(i = 0; cipher[i] != '\0'; i++)
                    printf("%02x ", (unsigned char)cipher[i]); // Hex output
                printf("\n");
                break;

            case 2:
                for(i = 0; cipher[i] != '\0'; i++)
                    plain[i] = cipher[i] ^ lp;
                plain[i] = '\0';
```

```
        printf("Decrypted text is: %s\n", plain);
        break;

    case 3:
        exit(0);

    default:
        printf("Invalid choice. Please try again.\n");
    }
}

return 0;
}
```

Output:

__Menu__

1:data encryption

2:data decryption

Experiment No. 9

DH Algorithm

Program in C:

```
#include <stdio.h>
#include <math.h>

long long int power(long long int a, long long int b, long long int P)
{
    if (b == 1)
        return a;
    else
        return ((long long int)pow(a, b)) % P;
}

int main()
{
    long long int P, G, x, a, y, b, ka, kb;

    P = 23;
    printf("The value of P : %lld\n", P);

    G = 9;
    printf("The value of G : %lld\n\n", G);

    a = 4;
    printf("The private key a for Alice : %lld\n", a);
    x = power(G, a, P);

    b = 3;
    printf("The private key b for Bob : %lld\n\n", b);
    y = power(G, b, P);

    ka = power(y, a, P);
    kb = power(x, b, P);

    printf("Secret key for the Alice is : %lld\n", ka);
    printf("Secret Key for the Bob is : %
```

Output:

The value of P : 23
The value of G : 9

The private key a for Alice : 4
The private key b for Bob : 3

Secret key for the Alice is : 9
Secret Key for the Bob is : 9

